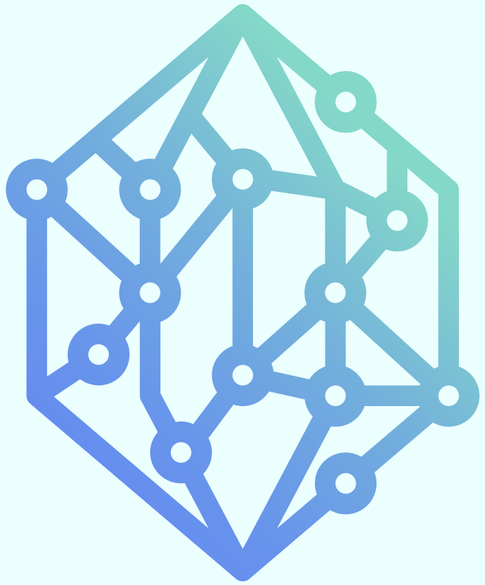


Project Title: Non-Invasive Glucose Monitoring via Gold Nanoparticle (AuNP) Surface Resonance

Name: Atreyi Barma Majumder

Registration No: 25BOE10040

Introduction:

Traditional glucose monitoring requires painful finger-pricking. This project explores a "Smart Patch" that uses nanotechnology to detect glucose levels in interstitial fluid (ISL) using optical sensors, providing a painless, real-time alternative for diabetic care.

Problem Statement:

Current invasive methods lead to low patient compliance and "needle phobia." There is a critical need for a high-sensitivity, non-invasive system that can detect minute glucose fluctuations without breaking the skin barrier.

Functional Requirements:

- Sensitivity: Must detect glucose concentrations between 30 to 400 mg/dL.
- Real-time Sync: Data must transfer to a mobile app via Bluetooth every 5 minutes.
- Alert System: Visual and haptic alerts for hypoglycemia or hyperglycemia.

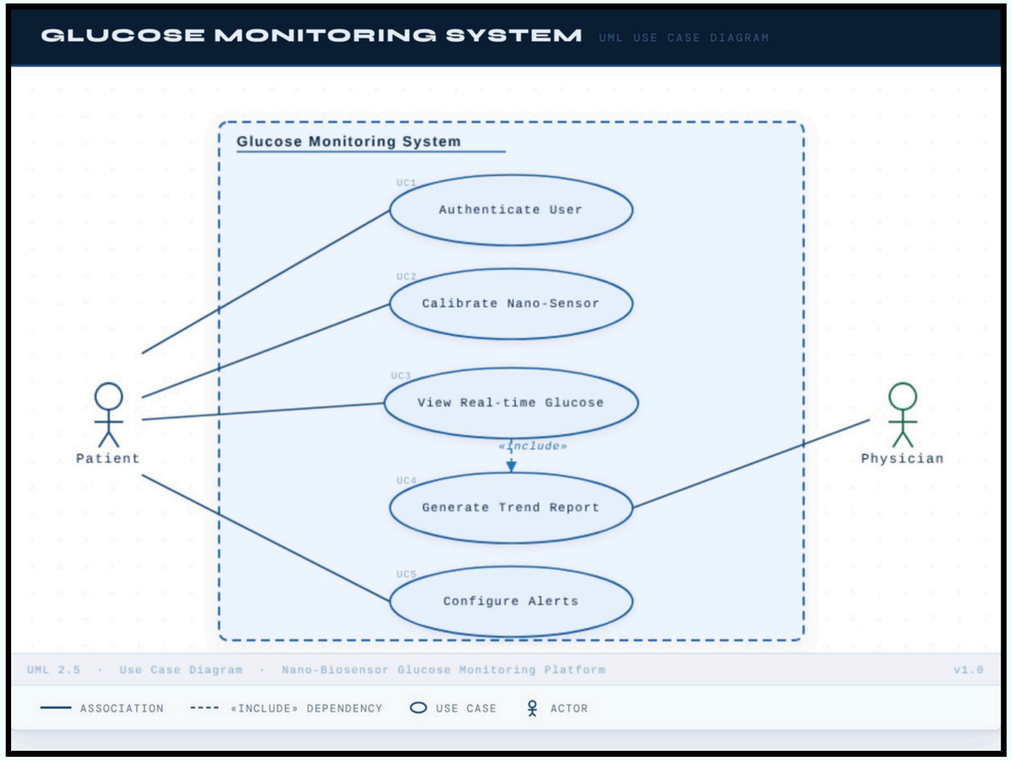
Non-functional Requirements:

- Biocompatibility: The patch adhesive must be non-irritating for 72-hour wear.
- Accuracy: Must maintain a <10% Mean Absolute Relative Difference (MARD) compared to blood tests.
- Battery Life: Must operate for 7 days on a single micro-battery charge.

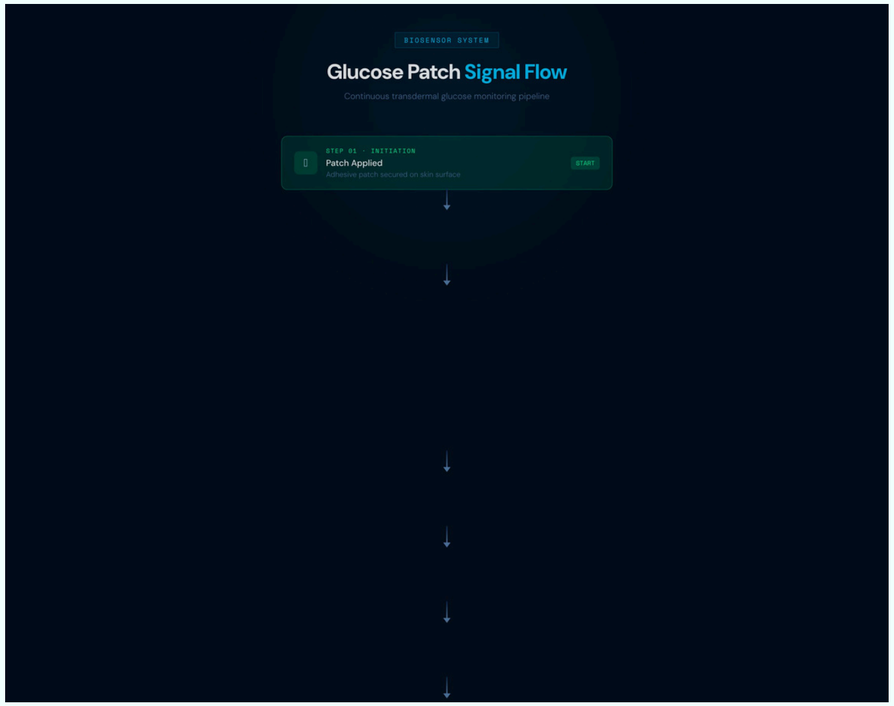
System Architecture:

- The system consists of three layers:
- Sensing Layer: Functionalized Gold Nanoparticles.
- Transducer Layer: Photodetector and Micro-LED.
- Processing Layer: ARM Cortex-M micro-controller and Bluetooth Low Energy (BLE) module.

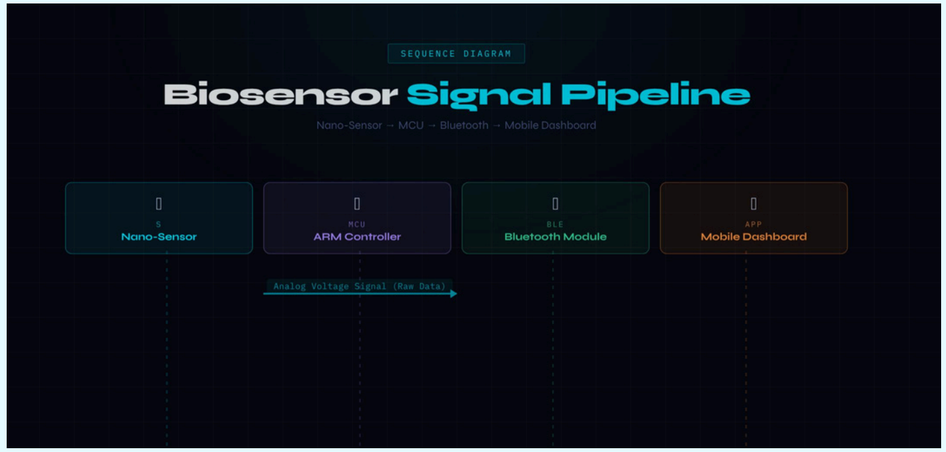
Design Diagrams:



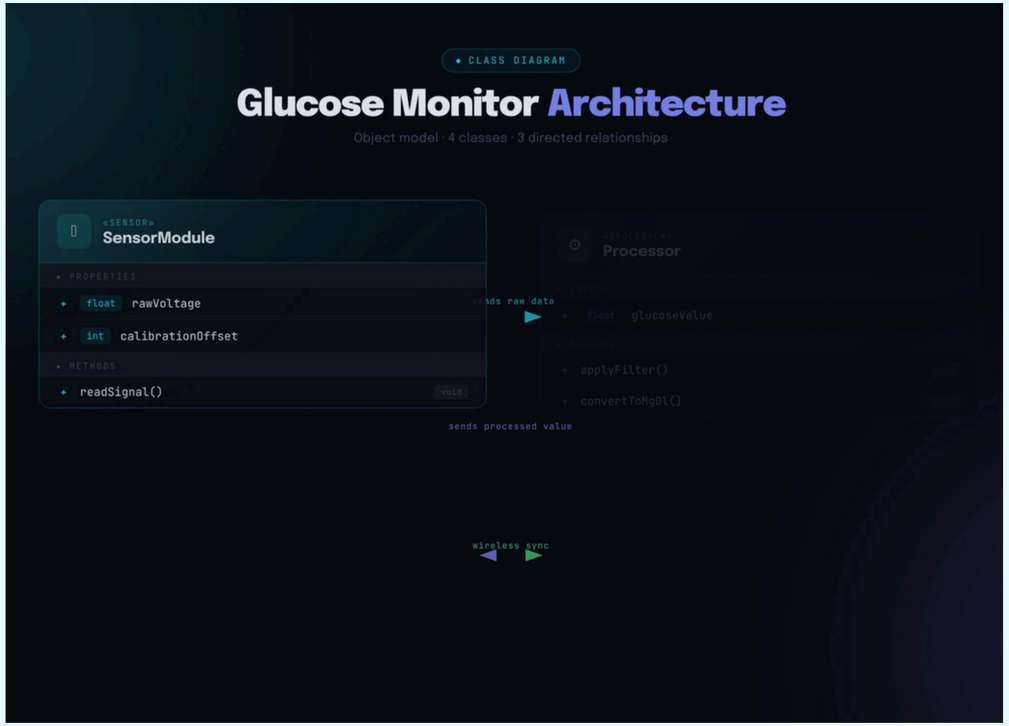
Use Case diagrams



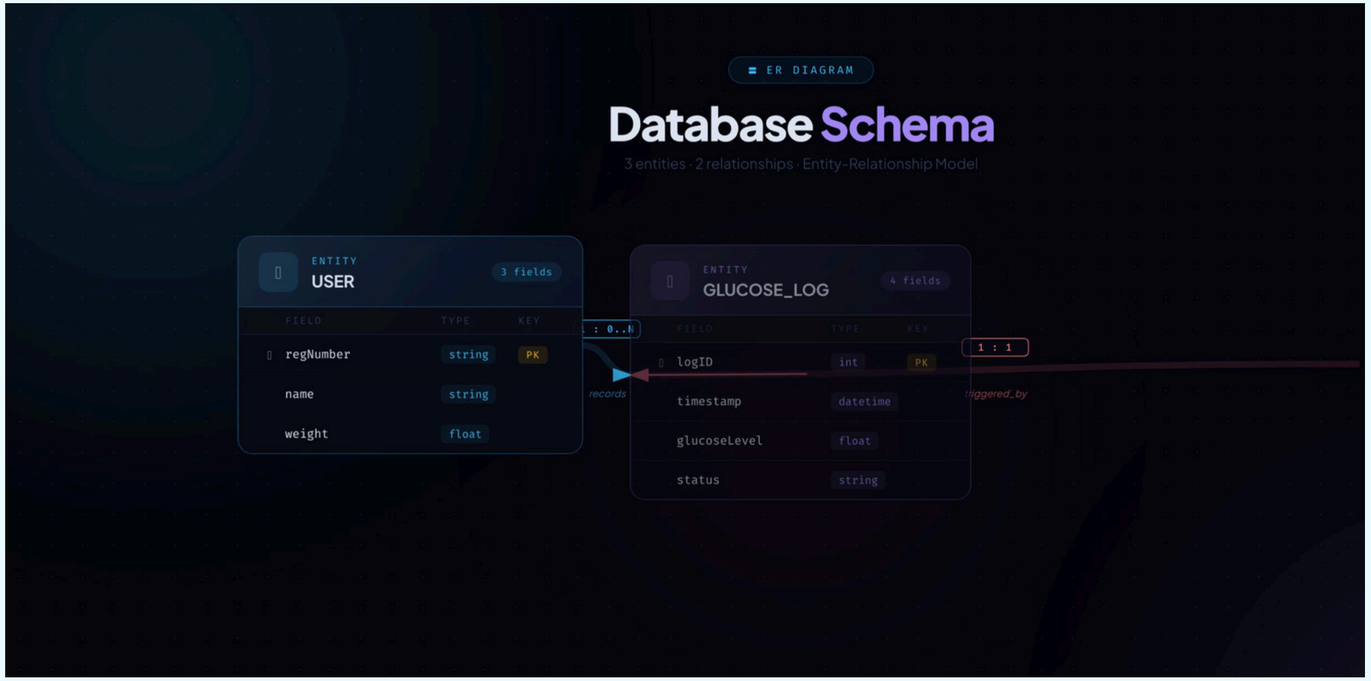
Workflow diagram



Sequence Diagram



Class diagram


ER DIAGRAM
Database Schema
3 entities · 2 relationships · Entity-Relationship Model
ENTITY
USER
3 fields
FIELD
TYPE
KEY
regNumber
string
PK
name
string
weight
float
records
ENTITY
GLUCOSE_LOG
4 fields
FIELD
TYPE
KEY
logID
int
PK
timestamp
datetime
glucoseLevel
float
status
string
1 : 1
triggered_by

ER Diagram

Design Decisions & Rationale:

- Decision: Using Gold Nanoparticles over Silver.
- Rationale: Gold offers superior chemical stability and better Surface Plasmon Resonance (SPR) properties, ensuring the device doesn't degrade when exposed to sweat.

Implementation Details:

- Hardware: Arduino Nano 33 BLE, high-sensitivity phototransistors, and a customized hydrogel patch.
- Software: Developed in C++ (firmware) and Flutter (mobile app) for cross-platform compatibility.

Screenshots / Results:

Testing Approach:

- Unit Testing: Validating the ADC (Analog-to-Digital) conversion accuracy.
- Integration Testing: Ensuring the BLE handshake remains stable within a 5-meter radius.
- Validation: Testing against a standard laboratory glucometer using synthetic interstitial fluid.

Challenges Faced:

- Signal Noise: Movement of the user created "motion artifacts" in the optical data.
- Calibration: Compensating for temperature changes which affect nanoparticle sensitivity.

Learnings & Key Takeaways:

- Gained expertise in integrating nanotechnology with embedded systems.
- Understood the importance of signal processing filters (Kalman filters) in medical sensors.

Future Enhancements:

- Integration with Insulin Pumps for a "Closed-Loop" artificial pancreas system.
- Energy harvesting from body heat to eliminate the need for batteries.

References:

- Nanotechnology in Medicine, Vol. 12, IEEE Press.
- Smith, J. (2025). Principles of Optical Biosensors.
- VIT Library Resources on Nanoparticle Synthesis.

THANK YOU!